

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Maximum - Minimum IV: On a closed bounded domain

Dr Francois Ebobisse Bille

We are interested here in finding the the points of maximum/minimum for a continuously differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ on a closed bounded domain D in $\mathbb{R}^n$. First of all the existence of such points follows from a higher-dimensional version of the Extremum Value Theorem (studied in first year Calculus), which states that, if $D \subset \mathbb{R}^n$ is closed (including its boundary) and bounded subset of $\mathbb{R}^n$, then a continuous function $f : D \rightarrow \mathbb{R}$ achieves maximum and minimum values.

We find the points of maximum and minimum of over D according to the following algorithm:

- (1) Find the critical points of the function of f lying inside the interior part of the domain D (if that interior is nonempty).
- (2) Find the points of maximum and minimum for the function f on the boundary of the domain D . Here we may use the method of the Lagrange multipliers.
- (3) Evaluate the function f at the points obtained in (1) and (2). The smallest of those values is the minimum and the corresponding points are points of minimum while the largest value is the maximum with its corresponding points being points of maximum.

Example 0.1 Find the points of maximum and the points of minimum for the function

$$f(x, y) := x - x^2 - y^2 \quad \text{over the set} \quad D := \{(x, y) \in \mathbb{R}^2 : x^2 + 4y^2 \leq 1\}.$$

Solution:

- (1) We find the critical points (x, y) of the function f such that $x^2 + 4y^2 < 1$. That is, we seek points (x, y) such that

$$\begin{cases} f_x(x, y) = 0, \\ f_y(x, y) = 0, \\ x^2 + 4y^2 < 1 \end{cases} \Leftrightarrow \begin{cases} 1 - 2x = 0, \\ -2y = 0, \\ x^2 + 4y^2 < 1 \end{cases}$$

We obtain the point $(1/2, 0)$ which clearly lies in the interior of the domain D .

- (2) Now, we find the points of maximum and minimum of the function f over the boundary of the domain D . This is equivalent to finding the points of maximum and minimum of the function f subject to the constraint $g(x, y) := x^2 + 4y^2 - 1 = 0$. To this aim, we use the method of the Lagrange multipliers. That is, we seek points $(x, y) \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$ such that

$$\begin{cases} 2x(1 + \lambda) = 1, \\ y(2 + 8\lambda) = 0 \\ x^2 + 4y^2 = 1 \end{cases}$$

From the second equation we get that $y = 0$ or $\lambda = -\frac{1}{4}$.

If $y = 0$, then we get from the last equation that $x = \pm 1$. So, we obtain the points $(-1, 0)$ and $(1, 0)$.

For $y \neq 0$ we get $\lambda = -\frac{1}{4}$ for which we get from the first equation that $x = \frac{2}{3}$ and hence, from the last equation that $y = \pm \frac{\sqrt{5}}{6}$. Thus, we get the points $\left(\frac{2}{3}, -\frac{\sqrt{5}}{6}\right)$ and $\left(\frac{2}{3}, \frac{\sqrt{5}}{6}\right)$.

(3) Now, we evaluate the function f at the points found in (1) and (2) to get

$$f(1/2, 0) = \frac{1}{4}; \quad f(-1, 0) = -2; \quad f(1, 0) = 0; \quad f\left(\frac{2}{3}, -\frac{\sqrt{5}}{6}\right) = f\left(\frac{2}{3}, \frac{\sqrt{5}}{6}\right) = \frac{2}{3} - \frac{4}{9} - \frac{5}{36} = \frac{1}{12} \approx 0.08.$$

Therefore, the function f achieves on the subset D its minimum value equal to -2 at the points $(-1, 0)$ and its maximum value equal to $\frac{1}{4}$ at the point $\left(\frac{1}{2}, 0\right)$.

Exercises 0.2 Find the points of maximum and/or minimum of each function subject to the given constraint.

- (a) $f(x, y) := x + y$ on the set $D := \{(x, y) \in \mathbb{R}^2: x^2 + 4y^2 \leq 16 \text{ and } y \leq 1\}$.
- (b) $f(x, y, z) := 2x + y^2 - z - 4$ on the set $D := \{(x, y, z) \in \mathbb{R}^3: x^2 + y^2 + z^2 \leq 1\}$
- (c) $f(x, y, z) := x^2 + (y - 1)^2 - 2z^2$ on the set $D := \{(x, y, z) \in \mathbb{R}^3: 2x^2 + y^2 + 3z^2 \leq 4\}$

The next set of notes on Multiple Integrals is ready!!